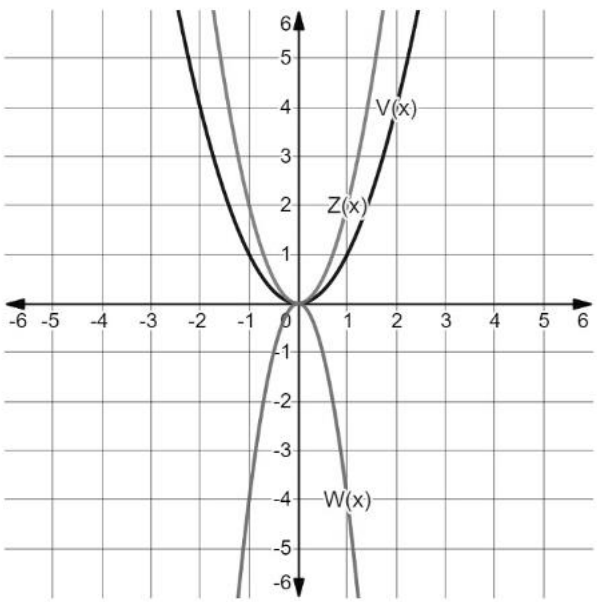
Algebra 1
Unit 13
Lesson 7 Practice

Name Answer Key
Date
Period

The functions $V(x)$, $W(x)$, and $Z(x)$ are shown on the graph, where $V(x) = x^2$, $W(x) = -4x^2$, and $Z(x) = 2x^2$. The table of values for each function are also shown.



$y = V(x)$		$y = W(x)$		$y = Z(x)$	
x	y	x	y	x	y
-3	9	-3	-36	-3	18
-2	4	-2	-16	-2	8
-1	1	-1	-4	-1	2
0	0	0	0	0	0
1	1	1	-4	1	2
2	4	2	-16	2	8
3	9	3	-36	3	18

Complete the statements below.

1. The coefficient of $V(x) = x^2$ is 1.
2. The coefficient of $W(x) = -4x^2$ is -4.
3. The y-values of $W(x)$ can be found by multiplying the y-values of $V(x)$ by -4.
4. The coefficient of $Z(x) = 2x^2$ is 2.
5. The y-values of $Z(x)$ can be found by multiplying the y-values of $V(x)$ by 2.

The table of values of the function $y = B(x)$ is shown.

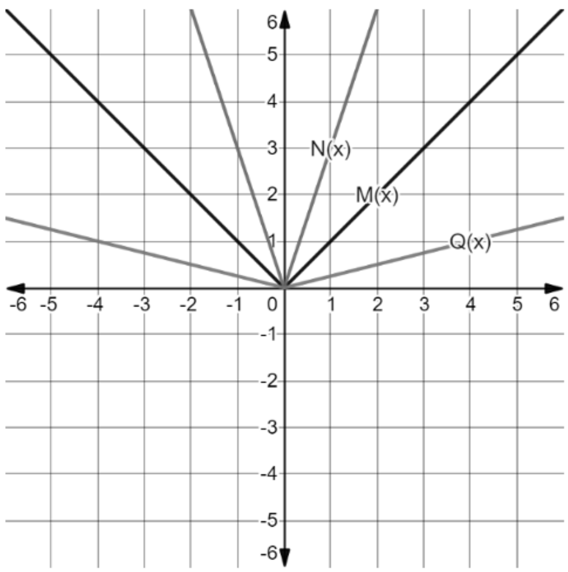
x	-6	-4	-2	0	2	4	6
y	4	7	10	13	16	19	22

6. Complete the sentence to describe the effect on the x-values for the function $y = B(2x)$. The x –values for $B(x)$ will be

○ multiplied
✓ divided

by 2.

The functions $M(x)$, $N(x)$, and $Q(x)$ are shown on the graph, where $M(x) = |x|$, $N(x) = |3x|$, and $Q(x) = |0.25x|$. The table of values for each function are also shown.



$y = M(x)$		$y = N(x)$		$y = Q(x)$	
x	y	x	y	x	y
-2	2	$-\frac{2}{3}$	2	-8	2
-1	1	$-\frac{1}{3}$	1	-4	1
0	0	0	0	0	0
1	1	$\frac{1}{3}$	1	1	4
2	2	$\frac{2}{3}$	2	2	8

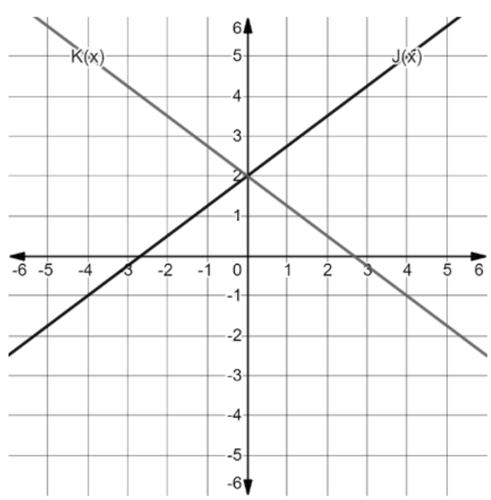
Complete the statements below.

7. The x-values of $N(x)$ can be found by [○ multiplying / ✓ dividing] the x-values of $M(x)$ by 3.

8. The x-values of $Q(x)$ can be found by [✓ multiplying / ○ dividing] the x-values of $M(x)$ by 4.

The functions $J(x)$ and $K(x)$ are shown on the graph where $K(x) = J(-x)$. The table of values for each function are also shown.

$y = J(x)$		$y = K(x)$	
x	y	x	y
-2	0.5	-2	3.5
-1	1.25	-1	2.75
0	2	0	2
1	2.75	1	1.25
2	3.5	2	0.5



9. What is the effect on the table of values when the x in $J(x)$ was replaced with $-x$ to create $K(x)$?

As the x-values increase, the y-values decrease in $k(x)$.

10. What is the effect on the graph when the x in $J(x)$ was replaced with $-x$ to create $K(x)$?

Reflection over the y-axis.